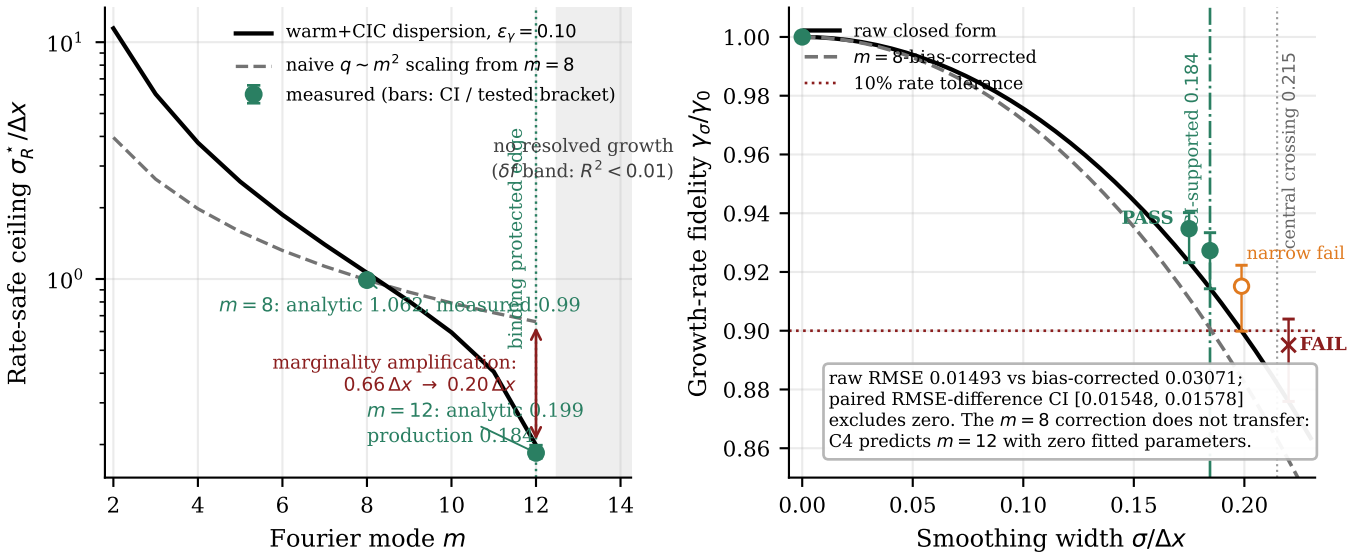
(a) Marginality tightens the safe width
warm+CIC dispersion, $\varepsilon_\gamma = 0.10$
naive $q \sim m^2$ scaling from $m = 8$
measured (bars: CI / tested bracket)
non-resolved growth
(δf band: $R^2 < 0.01$)
binning protected edge
$m = 8$: analytic 1.002, measured 0.99
marginality amplification:
$0.66\,\Delta x \rightarrow 0.20\,\Delta x$
$m = 12$: analytic 0.199
production 0.184
Rate-safe ceiling $\sigma_R^\star/\Delta x$
Fourier mode m
(b) Binding edge: linearized δf, $m = 12$
raw closed form
$m = 8$-bias-corrected
10% rate tolerance
PASS
1-supported 0.184
central crossing 0.215
narrow fail
FAIL
raw RMSE 0.01493 vs bias-corrected 0.03071;
paired RMSE-difference CI [0.01548, 0.01578]
excludes zero. The $m = 8$ correction does not transfer:
C4 predicts $m = 12$ with zero fitted parameters.
Growth-rate fidelity γ_σ/γ_0
Smoothing width $\sigma/\Delta x$